JI ZHOU

Master's Student in Robotics Engineering | Humanoid Robot Learning and Deployment

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RESEARCH PROFILE

Robotics master's student specialising in reinforcement-learning-based humanoid locomotion, imitation learning, sim-to-real deployment, and multi-robot systems. Hands-on experience across the complete development pipeline, including policy training, motion retargeting, ROS2/C++ deployment, motor communication, kinematics, system integration, and real-robot debugging. Current research interest: generating natural and physically feasible whole-body humanoid motions from high-level tasks and human demonstrations.

EDUCATION

Chongqing University | M.Eng. in Robotics Engineering (Embodied Intelligence) **2025 - Present**

Graduate research focused on humanoid robotics and embodied intelligence.

Chongqing University | B.Eng. in Mechatronics Engineering **2021 - 2025**

Training in mechanical design, embedded systems, control, sensing, and robotics.

RESEARCH INTERESTS

Humanoid Robot Learning	**Loco-Manipulation**	**Reinforcement and Imitation Learning**
Sim-to-Real Transfer	**Multi-Robot Coordination**	

SELECTED RESEARCH AND ENGINEERING EXPERIENCE

Full-Size Humanoid Robot: Reinforcement Learning and Sim-to-Real Deployment

Mar 2026 - Present

Primary contributor for policy training, deployment architecture, integration, and debugging

- Built an end-to-end locomotion pipeline in MJLab using PPO and an asymmetric Actor-Critic architecture; applied velocity and terrain curricula to improve policy generalisation.
- Introduced domain randomisation over robot mass, control latency, joint friction, foot-contact stiffness, and encoder bias to improve robustness and sim-to-real transfer.
- Developed a ROS2-based deployment framework for policy inference, command delivery, motor communication, action execution, observation feedback, and run-time data logging.
- Deployed the trained policy at 50 Hz using C++ and ONNX Runtime on the physical humanoid robot.
- Implemented Ethernet-to-CAN FD communication through UDP, Cannelloni, and SocketCAN for low-level control of DAMIAO actuators via DM-LinkX.
- Developed parallel-ankle kinematics to map virtual joint position, velocity, and torque to physical motors, together with test tools for motors, IMU, communication, trajectory tracking, and diagnostics.
- Completed real-robot integration and debugging; the robot has achieved stable walking.

Multi-Humanoid Dance Training and Fleet Control for Eight Unitree G1 Robots

Sep 2025 - Feb 2026

Motion training, multi-robot system design, software development, and on-site deployment

- Performed dance motion retargeting and policy training for Unitree G1 humanoid robots using GMR and BeyondMimic.
- Designed and implemented a UDP-based fleet-control system for eight robots, supporting fleet-level, group-level, and individual control.
- Implemented stage entrance and exit sequencing, synchronised dance triggering, run-time policy switching, robot status monitoring, and a graphical operator interface.
- Used timestamp-based synchronisation and fault-tolerant execution mechanisms to maintain coordinated motion under network fluctuation and temporary disconnection.
- Deployed the complete system in a live stage performance involving eight humanoid robots.

SELECTED ROBOTICS COMPETITION PROJECTS

National College Mechanical Innovation Competition — **National Second Prize** — **2024**

Project: Agricultural Scallion Transplanter

- Designed the vehicle electrical and control architecture, wrote microcontroller firmware, and selected motors, sensors, and structural components.
- Implemented chassis drive, motor control, serial and CAN communication, and remote-control functions.

China Robot Competition and RoboCup China Open — **National Second Prize** — **2023**

Project: Sorting Robot - Team Leader

- Led the mechanical design and fabrication of the robot body, manipulator, mobile base, and storage mechanism.
- Selected motors and sensors; developed STM32F407-based drive, vision-interface, control logic, and serial/I2C communication; used SolidWorks and Keil for design and embedded development.

National College Student Engineering Practice and Innovation Competition — **Third Prize, Chongqing Division** — **2022**

Project: Solar-Powered Electric Vehicle

- Contributed to vehicle mechanical design, transmission design, cam-based steering, and basic electrical integration.

TECHNICAL SKILLS

Programming	C++, Python, embedded C
Robot Learning	PPO, asymmetric Actor-Critic, imitation learning, motion retargeting, curriculum learning, domain randomisation
Robotics Software	ROS2, MJLab, GMR, BeyondMimic, ONNX Runtime
Communication	UDP, CAN, CAN FD, SocketCAN, Cannelloni, serial communication, I2C
Embedded and Hardware	STM32, actuator and sensor integration, IMU testing, motor debugging, real-time control
Mechanical	Robot mechanism design, parallel-ankle kinematics, SolidWorks, fabrication and system integration

HONORS AND AWARDS

2024 National Second Prize, National College Mechanical Innovation Competition

2023 National Second Prize, China Robot Competition and RoboCup China Open

2022 Third Prize, Chongqing Division, National College Student Engineering Practice and Innovation Competition

Second-Class Scholarship, Chongqing University

LANGUAGES

Chinese: Native communication. **English:** CET-6, score 521; capable of reading technical literature, technical writing, and research